Computer Vision–Informed Agent-Based Modeling for Visitor Behavior Analysis and Lead Qualification in Retail Environments

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ABSTRACT

Retail video analytics and agent-based simulation are usually studied separately, so observed visitor behavior rarely informs simulated sales operations. We present a prototype that connects the two. A computer vision pipeline detects people with a COCO-pretrained YOLO11n model, tracks them with ByteTrack, and measures dwell time in two floor zones. On a 142-second overhead clip from the MERL Shopping Dataset, it produced 8 zone-level records for 5 tracked identities. These records become arrival data and visitor features for a NetLogo model with visitor, lead, and sales-representative agents. In the model, a dwell-based interest score and simulated browsing behavior set the probability that a visitor becomes a lead, and three queue policies, first-come first-served (FCFS), SCORE, and SCORE+INTENT, decide which lead a representative serves next. A complete policy comparison has not yet been established. Identity fragmentation by the tracker limits the visitor counts, and the simulation is experimental: it does not predict real purchases.

Keywords: computer vision, agent-based modeling, visitor tracking, dwell time, lead qualification, NetLogo, retail analytics.

1. INTRODUCTION

Shops want to know which visitors deserve a salesperson's time. The traditional answer is manual observation or simple door counts, which say little about where visitors go or how long they stay. Studies of in-store paths show how much information movement data can hold [1], and modern detectors and trackers make it possible to collect such data from ordinary video [2], [3].

A measurement alone does not answer the operational question, however. A visitor who lingers at a shelf may be interested, but the business still has to decide which waiting visitor to serve first, how many representatives to schedule, and what happens to leads who wait too long. Agent-based modeling (ABM) is a natural tool for these questions because it represents individual visitors, representatives, and queues and lets their interactions produce system-level outcomes [4]. NetLogo is a widely used environment for building such models [5].

The gap we address is practical. In the work we reviewed (Section 2), video analysis stops at counts, paths, or actions, while simulation models of service operations assume their visitor behavior instead of taking it from video. We are not aware of a prototype that passes tracker-derived visitor features into a lead-qualification and queue model, but a systematic search is still needed before this can be claimed.

We therefore built a prototype that links the two parts. Figure 1 shows the two floor zones, drawn on the frame of an overhead clip from the MERL Shopping Dataset [6], in which dwell time is measured. The pipeline extracts first-seen time, last-seen time, zone dwell time, and a dwell-based interest score for each tracked visitor. These values are written to CSV files and read by a NetLogo model, where they influence whether a visitor becomes a lead and how leads are prioritized.

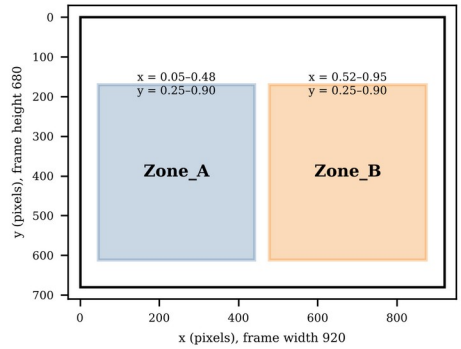


Fig. 1. Schematic of the two zones (Zone_A, left; Zone_B, right) used to measure dwell time, drawn to scale on the 920 × 680 pixel frame.

This work does not claim to detect real purchase intent or to predict real sales. Its purpose is to demonstrate and document a data-informed link between video observations and simulated retail operations.

Contributions.

- **Video-based visitor features.** A pipeline that turns retail video into visitor tracks, zone dwell time, and a dwell-based interest score, with input validation and an explicit account of its limitations.
- **A video-to-simulation interface.** A documented data format (arrival counts per minute and one feature record per visitor) that carries the video measurements into a NetLogo model, with a stated rule for how the imported interest score is normalized and combined with simulated behavior.
- **A lead-handling model with three policies.** An agent-based model of visitors, leads, and representatives that compares FCFS, SCORE, and SCORE+INTENT under repeated seeded runs, with real-data replay and synthetic arrivals kept separate.

The rest of the paper reviews related work (Section 2), describes the method (Section 3), reports results and their status (Section 4), and concludes (Section 5).

2. Related Work

2.1 Detection and tracking

YOLO introduced single-stage detection that predicts boxes and classes in one pass [2]. Detectors trained on COCO [7] include a "person" class, which is all this work needs. We use YOLO11n, the smallest model of the Ultralytics YOLO11 family [8]. Its authors have not published a formal paper, so the software release is cited. Tracking links detections across frames. SORT does this with a Kalman filter and the Hungarian algorithm [9], DeepSORT adds an appearance metric [10], and ByteTrack also associates low-confidence boxes to keep identities through occlusion [3]. Tracking quality is normally reported with CLEAR MOT metrics [11] and identity-based metrics such as IDF1 [12].

2.2 Dwell time and retail behavior

RFID tags on shopping carts were used to study how shoppers move through a supermarket [1]; that study tracked the cart, not the person. The MERL Shopping Dataset contains overhead videos of people shopping from shelves in a laboratory store, and was built for fine-grained action detection [6]. Dwell time is a behavioral indicator, not proof of purchase intent, so we treat it as one feature rather than as a measure of interest.

TABLE I. Related work by theme.

Theme	Representative work	Limitation for this problem
Person detection	YOLO [2]; COCO [7]; Ultralytics YOLO11 [8]	Finds people in single frames; gives no identity or duration
Multi-object tracking	SORT [9]; DeepSORT [10]; ByteTrack [3]; CLEAR MOT and IDF1 metrics [11], [12]	Identity switches under occlusion or re-entry; benchmarks do not measure dwell-time error
Retail behavior data	Supermarket shopping paths [1]; MERL Shopping Dataset [6]	Paths and actions are analyzed, but not connected to a service-operations model
Agent-based modeling	Tutorials and methods [4], [13], [14]; NetLogo [5]; ODD protocol [15], [16]	Visitor behavior is usually assumed rather than taken from video
Queues and service policies	Queueing theory [17]	Analytical models assume arrival and service distributions

2.3 Agent-based modeling and service queues

Agent-based models describe a system through autonomous agents and their interactions [4], [13], [14]. The ODD protocol gives a standard way to describe such models so that they can be understood and reproduced [15], [16], and we follow its structure in Section 3.4. Queueing theory studies waiting lines analytically [17] but typically assumes fixed arrival and service distributions, which are exactly what a video-informed model tries to replace with observed data.

2.4 Gap

Existing work analyzes video or simulates service operations, but the two are rarely connected through an explicit data interface. Our prototype targets that interface. A broader review is needed before claiming that no comparable work exists.

3. Methodology

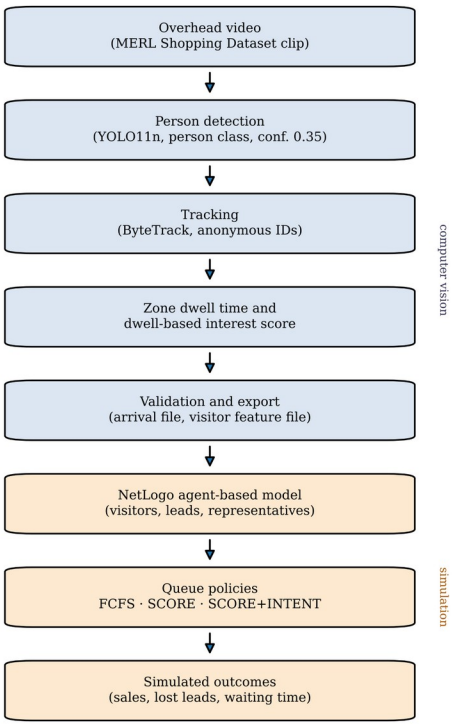


Fig. 2. System workflow from video to simulated lead and service outcomes.

3.1 Data

We use one clip from the MERL Shopping Dataset [6]: a fixed overhead camera in a lab space set up as a grocery shelf area. The dataset is released for non-commercial research, and its authors ask users to cite their paper.

3.2 Computer vision component

The detector is the COCO-pretrained Ultralytics YOLO11n model [8] with no fine-tuning. Every frame is processed with the person class only and a confidence threshold of 0.35. Detections are tracked with ByteTrack [3], using the default settings of the Ultralytics tracker. Each track receives an anonymous integer identity. No faces are recognized and no personal identity is stored.

TABLE II. Data and evaluation setup.

Item	Value
Resolution and frame rate	920 × 680 pixels, 30 frames per second
Length	4,274 frames (142.5 seconds)
Training, validation and test split	Not applicable: no model is trained or fine-tuned
Annotations used	None

Two rectangular zones are drawn as fractions of the frame (Fig. 1): Zone_A covers x = 0.05–0.48 and y = 0.25–0.90 (left side) and Zone_B covers x = 0.52–0.95 and y = 0.25–0.90 (right side). A frame counts toward a zone when the bottom-centre of a person's bounding box lies inside it. Dwell time is the number of such frames divided by the frame rate. The interest score is defined by the pipeline as

$$interest = min(100, round(dwell / 30\ s \times 100)).$$

This score is a deterministic function of dwell time. It is not an independent measurement, and it is not a measure of purchase intent. The pipeline writes one row per visitor and zone: visitor identity, first-seen time, last-seen time, zone, dwell seconds and interest score.

3.3 Data preparation and the video-to-simulation interface

A validation step reads the raw file and checks for a missing file, an empty file, missing columns, malformed rows, negative or non-numeric values, an interest score that is missing or above 100, and duplicate visitor-zone records. Two files are then produced:

- **Arrival file:** the number of unique visitors whose first-seen time falls in each minute of video. Several zone rows of the same visitor are counted once.
- **Visitor feature file:** one row per unique visitor with identity, first-seen time, maximum zone dwell (seconds) and maximum interest score (0–100).

3.4 Agent-based model (ODD-style summary)

Purpose. To compare lead-handling policies when sales representatives are limited, using visitor features that come from video.

Entities. Visitors walk through a simulated shop with two product zones and a counter, and carry simulated dwell times and, when available, an imported video record (identity, dwell, interest score). Leads are created from visitors, wait in a queue, and leave if they wait longer than a patience limit. Representatives serve one lead at a time.

Time and scale. One tick is one minute. Video times are in seconds and are converted to minutes only for reporting. Imported dwell is not used to set simulated dwell.

Arrival modes. In CV replay mode, arrivals follow the arrival file and the imported records are assigned in order of first-seen time. In synthetic mode, arrivals are Poisson with rate λ per minute and each visitor receives a record sampled with replacement from the video records, so the video features shape the synthetic population. The two modes are reported separately. If a visitor has no video record, only simulated behavior is used.

Interest and lead creation. When a visitor reaches the exit, simulated browsing gives a behavioral interest between 0 and 1 (points for dwelling at a product, visiting both products, returning, and reaching the counter, divided by 10). For visitors with an imported record, the imported score is divided by 100 to give a value between 0 and 1 (its defined scale), and the two values are averaged. The visitor becomes a lead with probability min(1, b + g × interest), where b and g are the baseline and gain parameters. Because the imported score is derived from dwell time, imported dwell itself is not used again anywhere in scoring, so the same information is not counted twice.

Lead quality and priority. A lead's hidden quality is min(1, 0.6 × interest + 0.4 × a random number). The score seen by the policy is the quality plus normally distributed noise (standard deviation s, clipped to 0–1), which stands in for an imperfect scoring model. A lead is "high-priority" when its hidden quality exceeds 0.7. Buying intent is a simulated yes/no attribute (30% of leads); it is not derived from video or text. The true purchase probability is min(1, 0.02 + 0.5 × quality, plus 0.1 if intent is present).

Policies. FCFS serves the lead that has waited longest. SCORE serves the lead with the highest observed score. SCORE+INTENT serves the lead with the highest observed score plus a bonus of 0.3 when intent is present.

Service. A representative serves a lead for a fixed service time. The lead then buys with probability min(1, skill × true purchase probability), where skill is drawn between 0.8 and 1.2 for each representative.

These values are modeling assumptions, not measurements.

3.5 Evaluation protocol

Simulation. Each policy is run for repeated seeded runs under each condition, with the same arrival mode, staffing and parameters for all policies. We report the mean and standard deviation of: visitors, leads generated, leads served, sales, lost leads, high-priority leads lost, conversion rate (sales divided by leads generated), and average wait (minutes, measured when service starts). FCFS is the baseline.

TABLE III. Simulation parameters.

Parameter	Value
Representatives	1 and 2
Patience (minutes)	10
Service time (minutes)	5
Baseline b and gain g	0.05 and 0.5
Score noise s	0.1
Intent bonus	0.3
Run length	480 ticks (synthetic) or until all video visitors are processed (CV replay)

Computer vision. Because identities in the clip can be checked against each other in time, we report how many identities overlap and the gaps between consecutive identities. Counting accuracy against a manual count, dwell-time error and tracking metrics (CLEAR MOT, IDF1) require annotation, which we do not have.

Reproducibility. Random seeds are recorded in every result row. All CSV files are generated by scripts from the video

and the raw CSV. Videos and outputs that show people are not redistributed.

4. Results and Discussion

4.1 Computer vision results

Table IV lists the eight zone-level records the pipeline produced from the clip. They belong to five tracked identities.

TABLE IV. Raw zone-level records (A = left zone, B = right zone).

Visitor	First seen (s)	Last seen (s)	Zone	Dwell (s)	Interest (0–100)
5	7.23	16.90	B	2.60	9
10	17.43	54.40	A	5.27	18
10	17.43	54.40	B	3.27	11
12	55.07	72.63	B	3.10	10
15	82.13	92.63	A	3.33	11
15	82.13	92.63	B	1.53	5
18	100.63	136.53	A	0.17	1
18	100.63	136.53	B	9.30	31

The arrival file has 3 visitors in minute 0 and 2 in minute 1. The feature file has one row per identity, with a maximum zone dwell of 2.6, 5.27, 3.1, 3.33 and 9.3 seconds and corresponding maximum interest scores of 9, 18, 10, 11 and 31.

Identity fragmentation. The five identities never overlap in time. The gaps between consecutive identities are 0.53 s, 0.67 s, 9.5 s and 8.0 s. Gaps of about half a second are typical of a tracker losing and reassigning the same person, and the longer gaps match periods when nobody is in view. In 11 frames sampled between 3 s and 130 s, at most one person is visible. Taken together, this suggests the clip shows a single shopper whose identity changed several times. Counts, arrival rates and per-visitor dwell from this clip should therefore be read as properties of the tracker's output and not of real visitors.

The zones also have a limit: the camera looks straight down, so the bottom-centre of a bounding box is only a rough stand-in for the position of the feet. The two rectangles partly cover the shelves and do not match product areas, so dwell is an experimental signal.

4.2 Simulation status

A complete policy comparison has not yet been established, so no policy results are reported here. The experimental protocol, with FCFS as the baseline, is described in Section 3.5.

CV replay. The replay of the real clip contains only a handful of visitors, so most runs produce few or no leads. We report it as an integration test that shows each stage of the pipeline works (visitor, lead, queue, service, outcome), and not as a policy comparison.

4.3 Limitations

- **Tracking quality.** Identities were fragmented in the only clip analyzed (Section 4.1), so visitor counts and per-visitor dwell are unreliable. Tracking metrics need frame-level ground truth.
- **Interest is dwell.** The interest score is a function of dwell time only. It does not measure intent, and it cannot be treated as independent evidence in addition to dwell.
- **One short clip.** A 142-second clip with few people supports a demonstration but not conclusions about arrival patterns or visitor behavior in general.
- **Staged setting.** The MERL clip was recorded in a laboratory arrangement of shelves, not in a working shop.
- **Model assumptions.** Purchase probabilities, patience, service time, intent and score noise are assumptions. The model has no learning, returning customers or staff fatigue.
- **No real-world claim.** Simulation results are experimental and do not show that any policy would improve a real business.
- **Privacy and licensing.** The clip shows a person. The method uses anonymous identities only, but a real deployment needs signage, consent rules and local legal review. The dataset is for non-commercial research.

5. Conclusion

We presented a prototype that carries visitor measurements from video into an agent-based model of leads and service policies. The pipeline produced dwell-based features from an overhead clip, and the model uses them to influence lead creation and prioritization under three queue policies. The main finding so far is a limitation: the tracker split what appears to be one shopper into five identities, so video-derived counts need validation before they drive a simulation. Next steps are manual annotation of the clip, tracker tuning on longer videos with several visitors, the seeded policy experiments with sensitivity analysis, and comparison with observed operations where available.

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